

A unified analysis of prosodic inversion and infixation

Peter Jenks and Hannah Sande
UC Berkeley

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Abstract

This paper argues that infixation and prosodic inversion are best analyzed using the same formal tool, namely, prosodic subcategorization. We show that second-position clitics that have been analyzed as prosodic inversion can be reanalyzed in Cophonologies by Phase with prosodic subcategorization frames, $\mathcal{P}$, that target prosodic constituents like ω , ϕ , ι . Infixes can be modeled in the same way, targeting sub-word prosodic constituents. This paper is the first in-depth exploration of the range of data that can be accounted for using the $\mathcal{P}$ -specifications in Cophonologies by Phase.

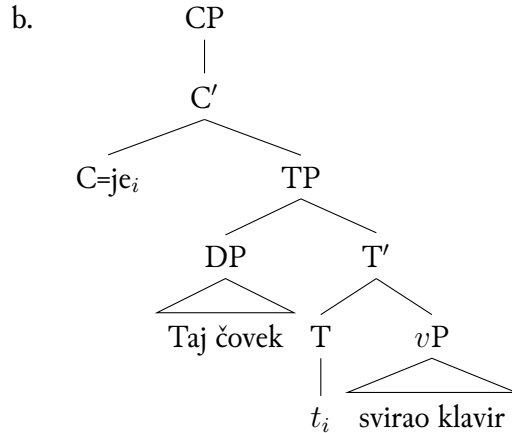
Keywords: clitics, second position, prosodic inversion, infixation, subcategorization, cophonologies by phase

1 What is prosodic inversion?

Prosodic inversion is the “inversion of a clitic and its host, such that the surface order of the two elements is the opposite of their syntactic order” (Halpern 1995: p. 2). An example of a Serbo-Croatian auxiliary analyzed as involving prosodic inversion is given in (1). This clitic is analyzed as syntactically higher than its surface position (1), but undergoing prosodic inversion to surface at the opposite edge of the immediately following prosodic word.

(1) Serbo-Croatian prosodic inversion

- a. *Taj =je čovek svirao klavir*
that AUX man played piano
‘That man played the piano’



While Halpern introduced this term in a technical sense, we will be using lowercase ‘prosodic inversion’ descriptively throughout this paper to describe cases where the surface position of a morpheme is different from its predicted syntactic position due to a prosodic requirement of that morpheme. More precisely, we will be using it to describe cases of words or morphemes ‘infixing’ to targets which are larger than a word, such as the Serbo-Croatian auxiliary *=je* above, essentially as the phrasal counterpart to the descriptive term ‘infixation.’

We observe, as did Anderson (2005), that clitics undergoing prosodic inversion very often have prosodically definable targets similar to the targets of infixation. More specifically, infixes are affixes which seek out edges of particular (sub-word) prosodic constituents (Kalin 2022, Yu 2007). For example, in Ulwa (Misumalpan) (2), the distributive infix (which reduplicates the following stressed CV sequence) targets the left edge of an iambic foot (Green 1999, Yu 2003).

- (2) Ulwa distributive infixation
- | | | | |
|----|---------|--------------------|-----------|
| a. | yám-ka | ‘good-ADJ’ | yayámka |
| b. | pí:-ka | ‘extinguished-ADJ’ | pipí:ka |
| c. | ihír-ka | ‘erect-ADJ’ | ihihírka |
| d. | bisí-ka | ‘small-ADJ’ | bisisí:ka |

In KiChaga (Bantu), the intensive infix is an /-n-/ that surfaces to the left of the final syllable (3) (Yu 2003: citing Inkelas, p.c.).

- (3) KiChaga intensive infixation
- | | | | |
|----|-----------|---------|----------|
| a. | lyi.an.gu | ‘light’ | lyianngu |
| b. | mu.i.li | ‘white’ | muinli |
| c. | mu.i.u | ‘black’ | muinu |

In Mlabri (Mon-Khmer), a nominalizing infix targets the first consonantal segment (Rischel 1995: p. 85). It surfaces as [-rn-] after the first consonant if the onset of the first syllable is a single C (4a,b), but as simply [-r-] if the onset of the first syllable is a consonant cluster (4c,d).

- (4) Mlabri nominalizing infixation
- | | | | | |
|----|-------|----------------------|---------|---------------------------|
| a. | kap | ‘to sing’ | krnap | ‘singing, song’ |
| b. | peelh | ‘to sweep the floor’ | prneelh | ‘a broom’ |
| c. | pluut | ‘to peel’ | prluut | ‘layer’ |
| d. | klaap | ‘to hold’ | krlaap | ‘forceps of split bamboo’ |

In all of these examples, infixes target edges of prosodic constituents: a stressed vowel in Ulwa, the final syllable in KiChaga, and the first consonant in Mlabri. In fact, clitics also target edges of prosodic constituents, as discussed by Anderson (2005), Inkelas (1990), Klavans (1985: a.o.).

In this paper we demonstrate that many well-known cases of cliticization qua prosodic inversion can be analyzed with the same mechanisms as infixation. The motivation for this unified approach comes from the shared formal profile of the two operations. First, both prosodic inversion and infixation seek the edge of a particular prosodic category, as long as we include C and V as prosodic categories. To satisfy this requirement, these operations can, but need not, break up syntactic domains (phases), and they can, but need not, break up phonological domains such as words and other prosodic constituents. These last two points are important because they represent parallel exceptions to well-known constraints on syntax: The Phase Impenetrability Condition (Chomsky 2001) restricts the ability of syntactic operations to access syntactic structures that have been transferred to PF, and analogously the Lexical Integrity Hypothesis (Bresnan & Mchombo 1995, Simpson 1983) says that syntactic operations cannot access boundaries internal to words.

While the observation that infixation and prosodic inversion bear some formal resemblance has been made before,¹ we show that the tools of Cophonologies by Phase (Sande 2019, Sande & Jenks 2018, Sande et al. 2020) can model the two in a unified way, via prosodic subcategorization requirements which are satisfied during Vocabulary Insertion. Cophonologies by Phase included tools for modeling morpheme-specific prosody, but only minimal motivation for such tools was provided in earlier work; we begin to address that gap here. We show that the parallel approach to word- and phrase-level phonology which the architecture supplies not only successfully models but predicts a unified approach to infixation and cliticization using prosodic subcategorization, first proposed for this purpose by Inkelas (1990).

After briefly discussing previous approaches to prosodic inversion in §2, and the Cophonologies by Phase model in §3, we examine three case studies of prosodic inversion: German *aber* in §4, Serbo-Croatian second-position clitics in §5, and English expletive infixation in §6. Expletive infixation provides a critical argument for a unified treatment of infixation and prosodic inversion, we will argue, in that expletive infixes originate adjoined to major phrases, and prosodically invert to a word-internal position to satisfy their subcategorization requirements. In §7 we discuss some of the limitations of our account as well as its predictions about impossible prosodic inversion patterns and about the typology of prosodic inversion more generally.

2 Previous analyses of prosodic inversion

This section provides an overview of previous approaches to deriving the surface position of clitics that involve prosodic inversion. These previous approaches can be categorized into three main types: syntax-only, phonology-only, and interface approaches. We summarize each of these three in turn here, pointing out challenges for each type of approach along the way.

¹Anderson (2005) notes that infixes and prosodically inverted clitics share many properties, and models both using alignment constraints; but he crucially models infixation in the lexicon and clitics post-syntactically, denying that they could be modeled at a unified position in the grammar. Kalin (2022) notes that “the definition [of infixation] could be expanded [...] to include free morphemes and complex words that appear inside of their stems (like English’s so-called expletive infixation) or even inside of phrasal constituents (like phonological second-position elements),” but does not explicitly consider clitics or other cases of prosodic inversion in her infixation typology.

2.1 Syntactic approaches

Several purely syntactic approaches to clitic placement have been proposed, e.g., Dimitrova-Vulchanova (1995) for Slavic second-position clitics. Arguments against purely syntactic approaches to clitic placement have been made time and again; see Anderson (2005), Bošković (2000), Halpern (1995) for several such arguments. One main argument against syntax-only approaches to clitic position in prosodic inversion is the following: deriving the correct surface position of clitics requires reference to phonological words and phrases, which are arguably not present in the syntax. It may well be the case that some items referred to as clitics should be handled purely syntactically, while others—any that require reference to phonological units—should be modeled another way. We are specifically interested in those clitics whose position is sensitive to phonological units.

2.2 Phonological approaches

Phonological approaches derive clitic placement via purely phonological mechanisms, arguing that the surface position of clitics in prosodic inversion is phonologically optimizing. See, e.g., Radanovic-Kocic (1988) on Serbo-Croatian and Bennett et al. (2016) on Irish. These approaches typically rely on an interface constraint specific to phonologically light elements which prefers them to be in non-prominent positions.

However, as we will see, not all clitic placement is phonologically optimizing. Additionally, clitic placement is often morphosyntactically sensitive, as argued by, e.g., Anderson (2005), Brown & Davis (to appear), Weisser (2024). Together, these facts make a purely phonological approach to prosodic inversion untenable as a generalized account of the phenomena.

2.3 Morphological or interface approaches

There are several approaches to clitic placement in prosodic inversion that rely on crucial interactions at the syntax/phonology interface, or in the morphological module in an approach where morphology applies after syntax but before phonology.

One such morphological approach involves specific morphological rules in Distributed Morphology. Marantz (1988) proposes the rule of Morphological Merger for this purpose, which, when it applies after vocabulary insertion, is called Local Dislocation by Embick & Noyer (2001). Both of these are post-syntactic operations that move a morpheme to an adjacent position after Vocabulary Insertion has occurred. Embick & Noyer describe this operation as one which “‘exchanges’ a structural relation between two elements at one level of representation for a different structural relation between the ‘heads’ of these elements at a subsequent level” (p. 561). While Embick & Noyer only define Local Dislocation for cases where a head inverts to the opposite side of an adjacent word, we find many cases of cliticization to a phrase across languages (for instance, the German facts in §2.4), or cases where morphemes cross several words, as with English expletive infixation described in §6. Embick and Noyer explicitly state that Halpern’s rule of Prosodic Inversion could in fact be seen as an instance of Local Dislocation (fn. 6, p. 561), an intuition we try to capture here. In many ways our approach is a generalization of Embick and Noyer’s approach, where the triggering feature for Local Dislocation/Prosodic Inversion is the presence of a prosodic subcategorization feature on the relevant morpheme.

A completely different interface-based approach would be to abandon the strict modularity of morphology and phonology and use a combination of morphological and phonological constraints to derive clitic placement. Approaches to clitic placement which adopt a ‘single-grammar’ model of these phenomena, such as Anderson (2005) or Werle (2009), require morphological indexing of constraints in order to capture the prosodic requirements of particular clitics. Such single-grammar approaches generally predict that clitic placement might be overridden for the purposes of phonological optimization. While it might be possible to find language-by-language justifications for the position of a particular clitic, there are clearly places of clitic placement which seem to be lexically arbitrary and non-optimizing. By treating prosodic inversion as a morphological phenomenon, such predictions are avoided. Furthermore, a theory which does not rely on indexed constraints adheres to strict modularity and is arguably simpler.

Another interface-based alternative involves prosodic subcategorization requirements associated with particular lexical items (Bennett et al. 2018, Bye 2007, Chung 2003, Halpern 1995, Herd 2003, Inkelas 1990, Paster 2006, Peperkamp et al. 1997, Weisser 2020, Yu 2003, Zec & Inkelas 1990). This seems to be the most promising direction for an analysis of prosodic inversion, and is the approach we will adopt.

Inkelas (1990) introduces the notion of prosodic subcategorization. Clitics for Inkelas are affixes which lack morphological selectional requirements (for her, a morphological categorization frame), and so do not attach to a word in the lexicon. Instead, clitics possess a prosodic subcategorization frame, and so attach to a word or phrase post-lexically. Halpern (1995) adopts the basics of the Inkelas (1990) approach, with a further stipulation that clitics syntactically move to the left edge of the word that they ultimately prosodically invert with.

Similar to us, Herd (2003) proposes that lexically specified prosodic subcategorization requirements are satisfied at the point of Vocabulary Insertion (in Distributed Morphology). Prosodic inversion is implemented as a feature² that applies at the end of Vocabulary Insertion. This approach is also similar to that of Weisser (2024), who proposes an Integration feature which applies at either the point of spell-out, resulting in inversion with a syntactic phrase, or at Vocabulary Insertion, resulting in prosodic inversion.³ We follow the spirit of these proposals in adopting the idea that prosodic inversion applies during Vocabulary Insertion and linearization, but we pursue the idea that it is the morpheme-specific prosodic subcategorization requirement itself that triggers prosodic inversion.

2.4 Problems for previous approaches

To illustrate problems that arise for the specific approaches to prosodic inversion described throughout this section, consider the contrastive coordinator *aber* ‘but’ in German, whose behavior is described in Weisser (2020, 2024).

In German, *aber* can appear either between two conjuncts (its original syntactic position), or it can prosodically invert with the following prosodic phrase (into the second conjunct), to which it suffixes (5):

²It is not clear in Herd (2003) what exactly this ‘feature’ is, or what its status in the grammatical system is.

³Weisser’s idea that morphological reordering can take place either before or after vocabulary insertion is shared with Embick & Noyer (2001); we are agnostic as to whether these are the same kind of morphological operation, but note that sensitivity to prosodic category should only be possible after some phonological representations have been projected, hence, at or after vocabulary insertion.

(5) German *aber* surface positions

- a. [_{CP} *Sie will tanzen*], (*aber*) [_{CP} *ich will (aber) nach hause gehen*].
 she wants dance but I want but to home go
 ‘She wants to dance but I want to go home.’ (Ross 1967: p. 163)
- b. [_{CP} *Sie will tanzen*], [_{CP} *ICH aber will nach hause gehen*].
 she wants dance I but want to home go
 ‘She wants to dance but I want to go home.’ (Weisser 2020)

Crucially, prosodic inversion of *aber* into the second conjunct is blind to syntactic constituency; its target is a prosodic phrase. Such generalizations are famously problematic for syntax-only approaches based on the standard assumption that prosodic constituency is only inserted post-syntactically (Inkelas & Zec 1990: a.o.).

We adopt the conclusion of Weisser (2020) that “German *aber* can undergo prosodically determined dislocation to a position after the phonological phrase to the right of its syntactic base-position.” Furthermore, the fact that *aber* targets a prosodic phrase, rather than a prosodic word, means that if an Embick & Noyer-style Local Dislocation account is adopted, it must be generalized to be able to dislocate words to positions after phrases.

Importantly, prosodic inversion is lexically-specific to *aber*, not derivable from the general behavior of conjuncts or phonology. Thus, other phonologically similar conjunctions do not show this same cliticizing behavior in German: *und* and *doch* cannot shift; *jedoch* must shift. This lexically specific behavior within members of the class of conjunctions shows that prosodic inversion behavior is due to a lexically specific property of these conjunctions, meaning that a purely phonologically optimizing approach is impossible (at least without reference to specific morphemes, at which point it becomes an interface-based approach).⁴

Our analysis closely resembles other existing proposals to clitic placement and prosodic inversion, including the recent proposal of Weisser (2024), but with some key differences. We assume that prosodic inversion is characterized by the properties in (6).

- (6) a. Prosodic inversion is triggered by lexically-specific prosodic subcategorization requirements (Inkelas 1990: a.o.)
- b. Exponent insertion occurs after exponent choice but before phonology (Kalin 2022).
- c. Morphophonological evaluation applies cyclically, at each cycle of spell-out. (Sande & Jenks 2018, Sande et al. 2020: a.o.)
- d. A realizational (vs. lexicalist) model of morphology with late insertion of vocabulary items, as in Distributed Morphology, is necessary.

Under this view, infixes will amount to morphemes with subcategorization frames which are smaller than words. Clitics undergoing prosodic inversion are morphemes with subcategorization frames which are the size of a prosodic word or larger. These various design features can be neatly integrated in Cophonologies by Phase.

⁴Indexed constraints, such as in Anderson’s approach, are undesirable because they do not adhere to strict modularity.

3 The Model: Cophonologies by Phase

Sande (2019), Sande & Jenks (2018), Sande et al. (2020) introduce Cophonologies by Phase (CbP), a model that assumes phonology applies at the spell-out of syntactic phase boundaries, and each phase may be sensitive to slightly different phonological grammars depending on the morphemes introduced in that phase. CbP assumes a Minimalist syntax, with phase boundaries (C, Voice, D, and categorizing heads (n, v, a)) which send syntactic constituents to PF. It assumes that the output of syntax is the input to post-syntactic modules (morphology, phonology), and it assumes that Vocabulary items, in the Distributed Morphology sense, are inserted late in the derivation, after syntax (i.e., it is a realizational approach to morphology).

One of the main contributions of CbP, one which has been the focus of much of our work to this point, is answering how phonology knows which phonological grammar (or co-phonology) to apply in which grammatical context. CbP does this via by enriching the content of vocabulary items to contain not only the segmental morphological content but also a prosodic subcategorization frame and a morpheme-specific constraint weight (or ranking) adjustment.

- (7) *Vocabulary Item* =_{DEF} an association of a morphosyntactic feature set with a tuple $(\mathcal{F}, \mathcal{P}, \mathcal{R})$:
- a. Featural content ($\mathcal{F}$): Underlying phonological form, i.e., Prosodic, suprasegmental, and segmental features
 - b. Prosodic subcategorization ($\mathcal{P}$): Prosodic subcategorization frame
 - c. Cophonology ($\mathcal{R}$): A partial constraint ranking (or weighting), which overrides a default master constraint ranking (or weighting) (Anttila 2002, Inkelas & Zoll 2007).

While (7c) is explored in depth in Sande (2018, 2019, 2020) and Sande et al. (2020), this paper examines the effects of prosodic subcategorization in more detail (7b), serving to exemplify how CbP can be used to account for prosodic-structure building, in this case with a specific focus on clitics whose placement described as involving prosodic inversion.

CbP is well equipped to explore prosodic inversion in that i) inversion is lexically specific to particular clitics (because CbP assumes the existence of a morpheme-specific prosodic subcategorization frame) and ii) inversion can apply across word boundaries (because CbP assumes phonological evaluation at phase boundaries which may contain multiple words, rather than word-by-word).

First, consider how prosodic subcategorization can be used to model infixation in a case where different infixal allomorphs occur after the first consonant. Mlabri data, first introduced in (4) is expanded upon in (8). A nominalizing infix appears after the first consonant, with two allomorphs: an /-rn/ allomorph appears after the first consonant when the onset of the first syllable is a singleton (8a), but an /-r/ allomorph after the first consonant when the first onset contains consonant cluster (8b).⁵

- (8) Mlabri nominalization (Rischel 1995: p. 85, as cited in Yu 2007: p. 76)
- a. Elsewhere: -rn- after first C

⁵We set aside a third allomorph where -l- occurs if the initial CC contains a rhotic.

guh	‘to be ablaze’	grnuh	‘flames’
kap	‘to sing’	krnap	‘singing, song’
peelh	‘to sweep the floor’	prneelh	‘a broom’
tek	‘to hit’	trnek	‘a hammer’

b. Initial CC: *-r-* after first C

kwel	‘to be rolled up’	krwel	‘spiral’
glaʔ	‘to speak’	grlaʔ	‘speech, words’
pluut	‘to peel’	prluut	‘layer’
klaap	‘to hold’	krlaap	‘forceps of split bamboo’

Yu (2003) assumes infixes like the Mlabri nominalizer possess prosodic subcategorization frames; we incorporate this intuition into Cophonologies by Phase by including subcategorization frames as part of the $\mathcal{P}$ value in the lexical entry, where X represents the morpheme in question and n represents a nominalizing head in the syntax.

(9) Mlabri nominalization infixes

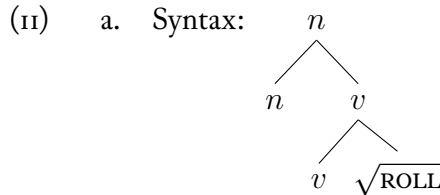
$$\begin{aligned}
 \text{a. } n &\longleftrightarrow \left\{ \begin{array}{l} \mathcal{F}_2 : /r/ \\ \mathcal{P}_2 : C-X \\ \mathcal{R}_2 : \emptyset \end{array} \right\} / _CC \\
 \text{b. } n &\longleftrightarrow \left\{ \begin{array}{l} \mathcal{F}_1 : /rn/ \\ \mathcal{P}_1 : C-X \\ \mathcal{R}_1 : \emptyset \end{array} \right\} / \textit{elsewhere}
 \end{aligned}$$

Note that the environment (which appears after the ‘/’) is a Condition on Insertion (=COIN), while the $\mathcal{P}$ is a Condition on Position (=COP), in the terms of Kalin & Rolle (2022). Hence, in this architecture, satisfying the COIN is a necessary precondition for accessing the COP by design.

For Kalin (2022), infixation is a derivational step which takes place during PF after VI and linearization, but before the phonology (p. 672), (10).

(10) linear concatenation $\rightarrow$ exponent choice $\rightarrow$ **infixation** $\rightarrow$ (morpho)phonology

Below we see the derivation of Mlabri *k-r-wel*, ‘spiral’. Concatenation is necessary so that phonologically-conditioned allomorphy, which we assume depends on phonological adjacency between the nominalizing head and the root, can apply. The Condition on Insertion, the environment, is satisfied during VI by the initial /kw/ sequence of the verb ‘roll’, resulting in the selection of the *-r-* nominalization infix. The Condition on Position, the subcategorization frame, is satisfied after VI, positioning *-r-* after the initial consonant in the root.



b. Linear concatenation: $n \frown v \frown \sqrt{\text{ROLL}}$ Linearly ordered features

- c. Exponent choice (=VI Part 1): $(\mathcal{F}_2, \mathcal{P}_2, \mathcal{R}_2)^\frown /kw\text{el}/$ COIN satisfied
- d. Exponent insertion (=VI Part 2, i.e. “infixation”): $/k\text{-r-w}\text{el}/$ COP satisfied

The Mlabri example in (11) involves two cycles of spell-out, each triggered by a categorizing head (Sande et al. 2020). This accounts for the ability of the n head to ‘see’ the phonology of the verb (12).

(12) Two-cycle infixation

	Cycle 1: The lexical verb	Cycle 2: Nominalization
Input: Syntax:	$[_v v \sqrt{\text{ROLL}}]$	$[_n n /kw\text{el}/]$
Step 1: Linear concatenation	$v^\frown \sqrt{\text{ROLL}}$	$n^\frown /kw\text{el}/$
Step 2: Exponent choice	$(v^\frown \sqrt{\text{ROLL}}) \leftrightarrow /kw\text{el}/$	$(n \leftrightarrow (\mathcal{F}_2, \mathcal{P}_2, \mathcal{R}_2))^\frown /kw\text{el}/$
Step 3: Exponent insertion	$/kw\text{el}/$	$/k\text{-r-w}\text{el}/$

Note that in Cycle 2, the phonological content of the verb is present; this is the relevant environment conditioning the choice of n among the two options in (9); their COIN. The resulting lexical entry includes the full tuple $(\mathcal{F}, \mathcal{P}, \mathcal{R})$. The content of $\mathcal{P}$ demands that the relevant exponent is infixed after the initial CC from the previous cycle, an instance of infixation.

Kalin argues that infixation is best understood as morphological, attributable to the sub-categorization requirements of a specific morpheme rather than narrow phonology, because it sometimes produces non-optimizing outcomes that require later repairs in the phonological component, and also because the output of infixation is often the input to later phonological changes. We preserve this intuition in our analysis of prosodic inversion, which we take to involve obligatory satisfaction of prosodic subcategorization requirements before phonology.

4 Case study 1: German *aber*

Now we will show how the model of infixation outlined in the previous section extends to prosodic inversion. German *aber* is instructive as a first case study: because it is a full phonological word, and not phonologically defective in any way, its phonologically-determined position must be due to some lexically-encoded constraint rather than general phonological principles (Weisser 2020; ex. 4, ex. 21, citing Ross 1967).

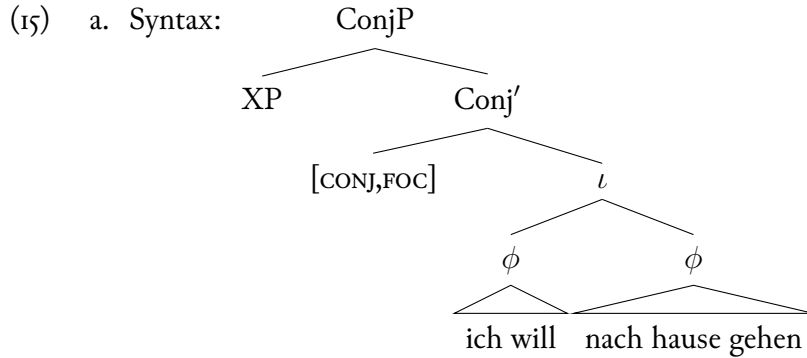
(13) German *aber* surface positions

- a. $[_{CP} \text{Sie will tanzen}], (\text{aber}) [_{CP} \text{ich will (aber) nach hause gehen}]$.
 she wants dance but I want but to home go
 ‘She wants to dance but I want to go home.’ (Ross 1967: p. 163)
- b. $[_{CP} \text{Sie will tanzen}], [_{CP} \text{ICH aber will nach hause gehen}]$.
 she wants dance I but want to home go
 ‘She wants to dance but I want to go home.’ (Weisser 2020)

We can capture the lexical requirement that *aber* be suffixed to a phonological phrase with the $\mathcal{P}$ component of its lexical entry, as with infixes:

$$(14) \quad [\text{CONJ}, \text{FOC}] \longleftrightarrow \left\{ \begin{array}{l} \mathcal{F} : \quad /[\text{aber}]_{\omega}/ \\ \mathcal{P} : \quad]_{\phi} \text{-X} \\ \mathcal{R} : \quad \emptyset \end{array} \right\}$$

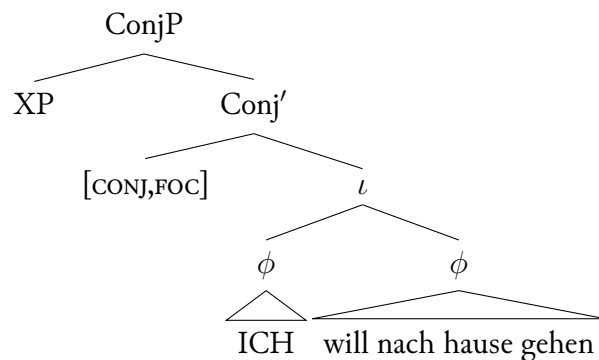
As with infixation, the prosodic positioning of *aber* occurs during exponent insertion. The derivation in (15) illustrates the steps. We begin at a point in the derivation where the CP complement of *aber* has been spelled out, assuming an asymmetric structure for coordination. Because it has been spelled out, it contains only layered prosodic information, including intonational phrase (ι) and phonological phrase (ϕ) boundaries, but not syntactic information (15a). Default concatenation rules linearize the focused conjunction head to the left of the intonational phrase which is its complement (15b). Following this step, Vocabulary Insertion applies, and *aber* is selected (15c). Part of its content is the subcategorization specified in $\mathcal{P}$, which demands that it occur at the right edge of ϕ . When exponent insertion applies, the $\mathcal{F}$ content is placed to the right of a phonologically adjacent ϕ (15d). This then derives prosodic inversion analogous with infixation.



- b. Linear concatenation: $[\text{CONJ}, \text{FOC}] \frown (\iota (\phi \text{ ich will }) (\phi \text{ nach hause gehen}))$
 c. Exponent choice (=VI): $([\text{CONJ}, \text{FOC}] \leftrightarrow [\omega \text{ aber}]) \frown (\iota (\phi \text{ ich will }) (\phi \text{ nach hause gehen}))$
 d. Exponent insertion (+infixation): $(\iota (\phi \text{ ich will }) \text{-aber } (\phi \text{ nach hause gehen}))$

Compare the following derivation where contrastively focused *ICH* attracts *aber*, as it forms a phonological phrase.

(16) a. Syntax:



- b. Linear concatenation: $[CONJ, FOC] \frown (\iota(\phi \text{ ICH})(\phi \text{ will nach hause gehen}))$
 c. Exponent choice (=VI): $([CONJ, FOC] \leftrightarrow [\omega \text{ aber}]) \frown (\iota(\phi \text{ ICH})(\phi \text{ will nach hause gehen}))$
 d. Exponent insertion (+infixation): $(\iota(\phi \text{ ICH})\text{-aber})(\phi \text{ will nach hause gehen})$

Thus, the surface location of *aber* is simply due to the phonological structure of its complement.

Weisser highlights the successful predictions of a ϕ -inversion story, including the ability of nested ϕ -phrases to produce multiple possible adjunction sites. As with infixation, $\mathcal{P}$ is satisfied upon exponent insertion, not as part of the phonology.

5 Case study 2: Serbo-Croatian second position clitics

We now turn to the case of second position clitics in Serbo-Croatian (SC), the focus of much earlier work.⁶ The similarities to German *aber* are striking; second-position clitics in Serbo-Croatian occur either after the first syntactic constituent (17a) or prosodic word (17b). The latter can result in the clitics intervening in a syntactic constituent as in Schütze (1994), Zec & Inkelas (1990: a.o).

- (17) a. $[_{DP} \text{ Taj } \check{c}ovek] =_{joj}=ga=je \text{ poklonio.}$
 that man =her=it=AUX presented.
 ‘That man presented her with it.’
 b. $[_{DP} \text{ Taj } =_{joj}=ga=je \text{ } \check{c}ovek] \text{ poklonio.}$
 that =her=it=AUX man presented.
 ‘That man presented her with it.’ (Zec & Inkelas 1990: p. 367)

These clitics can never be clause initial in SC (18)

- (18) a. $Ivan =ga=je \text{ } \check{c}esto \text{ } \check{c}itao$
 Ivan =it=AUX often read
 ‘Ivan often read it’
 b. * $ga=je \text{ } Ivan \text{ } \check{c}esto \text{ } \check{c}itao$
 it=AUX Ivan often read
 ‘Ivan often read it’ (Cavar & Wilder 1994: p. 25-6)

⁶Serbo-Croatian clitics, and Slavic clitics more generally, have been intensively studied. See Werle (2009) for a recent review of different approaches to Serbo-Croatian clitics and an OT alternative to a subcategorization approach for these cases.

Prosodically light elements such as weak prepositions cannot host the clitics (19).

- (19) a. *U kući =je Pétar.*
 In house =AUX Peter.
 ‘In the house is Peter.’
 b. **U =je kući Pétar.*
 In =AUX house Peter.
 ‘In the house is Peter.’ (Zec & Inkelas 1990: p. 367)

Contrasts like (20) provide evidence against movement-based approaches (Schütze 1994).⁷

- (20) a. [*U ovu*] =je [*veliku sobu*] ušao.
 in this is big room entered
 ‘He entered this big room.’
 b. ?*[*U ovu*] Jovan ulazi [*veliku sobu*].
 in this Jovan enters big room
 ‘Jovan is entering this big room.’ (Bošković 2001: p. 19)

In (20a), the clitic *=je* occurs between the determiner and adjective of a PP constituent. While Serbo-Croatian often allows adjectives or demonstratives to undergo left-branch extraction, the main alternative analysis of examples like (17b) with a split subject, examples like (20b) demonstrate that left-branch extraction is impossible for cases where a preposition and determiner are split from their complements. Hence, (20b) shows that a well-formed target of prosodic inversion often cannot undergo left-branch extraction, meaning that left-branch extraction undergenerates possible sites for clitics in Serbo-Croatian.

To summarize, then, second-position clitics in Serbo-Croatian either follow an initial XP or the initial ω . Our contribution to the Serbo-Croatian clitics literature is very modest: To show that a prosodic inversion analysis of Serbo-Croatian can be captured in our model, building on the account originally proposed by Inkelas (1990).

We follow earlier work in assuming that SC clitics are syntactically located in C (Bošković 2001, Cavar & Wilder 1994: i.a.), and, given their inflectional content, we assume they are the realization of a T head which has moved to C and serve as the exponent of both heads. The requirement that *=je* and other Serbo-Croatian clitics be preceded by a prosodic word can be encoded as its $\mathcal{P}$ -content.

(21) *Serbo-Croatian second-position clitics:*

$$[C, T, \text{PRES}, 3P, \text{SG}] \longleftrightarrow \begin{cases} \mathcal{F} : & /je/ \\ \mathcal{P} : &]_{\omega} X \\ \mathcal{R} : & \emptyset \end{cases}$$

We examine two main cases: XP=CL cases in (22a) and the constituent-busting X=CL cases (22b).

- (22) a. [_{DP} *Taj čovek*] =joj=ga=je poklonio.
 that man =her=it=is presented.
 ‘That man presented her with it.’

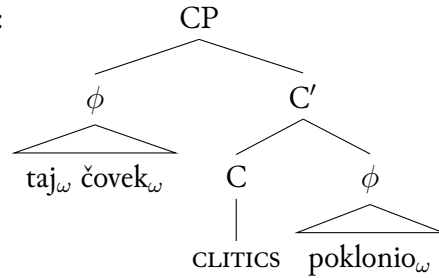
⁷Bošković (2001: 19–21) offers a spirited reply, arguing that, among other things, some equivalents to (20a) are degraded with longer clitic sequences.

- b. *Taj =joj=ga=je čovek poklonio.*
 that =her=it=is man presented.
 ‘That man presented her with it.’ (Zec & Inkelas 1990: p. 367)

If an XP occupies, [Spec, CP], C will be inserted in its surface position after ω .

The simple cases involve XPs which have moved to [Spec, CP] via A'-movement. In such cases, the syntax supplies the clitics in C with a suitable phonological host, and no displacement is needed. Note that we are assuming that the VP has been spelled out previously, here just the verb *poklonio* ‘presented,’ and also that the entire CP, including its specifier, is the relevant spell-out domain (23).

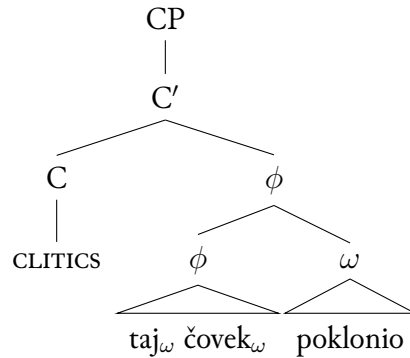
(23) a. Syntax:



- b. Linear concatenation: $(\phi \text{ taj}_\omega \text{ čovek}_\omega) \frown [\text{CLITICS}] \frown (\phi \text{ poklonio}_\omega)$
 c. Exponent choice (=VI): $(\phi \text{ taj}_\omega \text{ čovek}_\omega) \frown (\text{CLITICS} \leftrightarrow =\text{joj}=\text{ga}=\text{je}) \frown (\phi \text{ poklonio}_\omega)$
 d. Exponent insertion (+infixation): $(\phi \text{ taj}_\omega \text{ čovek}_\omega) =\text{joj}=\text{ga}=\text{je} (\phi \text{ poklonio}_\omega)$

Prosodic inversion applies when there is no XP in [Spec, CP]. Lacking a suitable phonological host to its left, exponent insertion results in the clitic group attaching to the DP-initial demonstrative *taj* ‘that’ (24)

(24) a. Syntax:



- b. Linear concatenation: $[\text{CLITICS}] \frown (\phi \text{ taj}_\omega \text{ čovek}_\omega) (\phi \text{ poklonio}_\omega)$
 c. Exponent choice (=VI): $(\text{CLITICS} \leftrightarrow =\text{joj}=\text{ga}=\text{je}) \frown (\phi \text{ taj}_\omega \text{ čovek}_\omega) (\phi \text{ poklonio}_\omega)$
 d. Exponent insertion (+infixation): $(\phi \text{ taj}_\omega =\text{joj}=\text{ga}=\text{je} \text{ čovek}_\omega) (\phi \text{ poklonio}_\omega)$

Prosodic inversion in both German and Serbo-Croatian shares a ‘minimal compliance’ flavor: they only apply when necessary, and then target the closest available host. Our current analysis captures the ‘only when necessary’ part of this generalization, as no prosodic inversion applies when subcategorization is satisfied in the input. However, we have not accounted for the observation that prosodic inversion targets the closest available host, ruling out an output like *taj čovek*

poklonio=joj=ga=je, where clitics attach to the end of the sentence. However, again the minimal compliance feature is shared between prosodic inversion and infixation: infixation targets word edges just as prosodic inversion targets the edge of prosodic phrases. We discuss some ways of deriving the shared minimal compliance generalization in Section 7.

6 Case study 3: English expletives

In our final case study we examine English expletive infixation (McCarthy 1982, Zingler 2024). We argue that this process can and should be modeled as prosodic inversion, and in fact that it straddles the boundaries of cliticization and infixation in a way which follows naturally from a unified account. We model expletive infixes as adverbs which prosodically invert to word-internal positions. Because expletives originate external to the words they occur inside of, they provide a crucial argument for a unified approach to infixation and prosodic inversion, as they constitute prosodic inversion to a word-internal position.

We begin with the well-known observation that many English expletives can appear word-internally, and are often thought of as infixes when they do so (25).

- (25) a. This is fan-fuckin'-tastic.
b. I'm going to Aca-fuckin'-pulco.

These word-internal expletives have well-known prosodic restrictions, namely that they cannot occur before unstressed syllables (26).

- (26) a. *This is fantas-fuckin'-tic.
b. *I'm going to Acapul-fuckin'-co.

McCarthy (1982) proposes a rule of English expletive infixation stipulating that they 'lodge' (procliticize) to the left edge of a foot, including to ω -internal positions ($=\Sigma$):

- (27) Expletive Infixation (Metrical Version) (McCarthy 1982: p. 578)
 $X [Y]_{\Sigma}$
 $I \quad 2_{\Sigma} \rightarrow I \text{ EXPLETIVE } 2_{\Sigma}$

While McCarthy and earlier discussions of English (Aronoff 1976, Siegel 1974) describe this process as infixation, Zingler (2024) points out that this process lacks properties of canonical infixation, or affixation more generally. First, Zingler points out that expletive 'infixes'—a term we will continue to use for convenience—are often morphologically complex, like *fuck-ing* itself or also British *blood-y*, which can similarly infix; affixes are, by definition, single morphemes. Second, expletive insertion does not select for a particular lexical category of its host, as it is possible in adverbs, as in *abso-bloody-lutely*, or nouns, as in, *kanga-bloody-roo*. Finally, the optionality in the positioning of expletives is unlike (at least some) infixation patterns. The existence of pairs of grammatical infix placements were first noticed by McCarthy (28).

- (28) Optional expletive placement (McCarthy 1982: p. 580)
a. (,Ta.ta)-fu.ckin-ma('gou.che) or (,Ta.ta.ma)-fu.ckin-('gou.che)
b. (,Win.ni)-fu.ckin-pe('sau.kee) or (,Win.ni.pe)-fu.ckin-('sau.kee)

As the parentheses above suggest, these pairs arise whenever there is a medial unstressed syllable which can be parsed with either with the maximal foot to its left or right (or which perhaps remains unparsed).

To Zingler's observations above, we would add that, like German *aber*, *fuckin'* and other expletive infixes are not prosodically deficient: they bear stress and hence are phonological words on their own. This suggests that the infixation process is not motivated by general phonological constraints.

We turn now to our analysis of expletive infixation as a kind of prosodic inversion. A starting point for our analysis is that it is desirable to unify an analysis of infixed expletives with the use of these expressions when they are not infixed. Just as *aber* and Serbo-Croatian clitics alternate between syntactically derivable locations—as heads with filled specifiers, for example—expletives like *fuckin'* when they are not infixed have the same general syntactic distribution as focus-associating adverbs like *only* or *even*. These adverbs have a well-known distribution: they must attach to major constituents, which are often labeled with a system of F(ocus)-marking, following the foundational work of Rooth (1985). But these adverbs are quite free in their distribution, attaching to DPs, VPs, and PPs (Toosarvandani 2010).

- (29) a. Only [MAX]_F has made sushi.
 b. Carrie has only visited [the Golden Gate BRIDGE]_F.
 c. My mother will walk only [to WORK]_F. (Toosarvandani 2010: p. 6)

Furthermore, while English focus-associating adverbs can attach to embedded CPs (30), they resist attaching to root clauses (31)

- (30) a. ...it said only [that Rushton was being held]_F... (example from literature)
 b. What it said was only [that Rushton was being held]_F.
 c. Only [that Rushton was being held]_F, it said. (Toosarvandani 2010: p. 7-8)
- (31) a. # Only Max has made [SUshi]_F.
 b. Max has (only) made [SUshi]_F.
 c. Only [MAX]_F has made [SUshi]_F. (Toosarvandani 2010: p. 6)

What is important about (31) is that sentence-initial *only* cannot associate the object under c-command, although focus on the object in the absence of the sentence-initial adverb is otherwise licit. A focused subject can directly associate with *only* here as it is adjoined to, hence c-commands, the subject DP.

Expletives have a similar distribution; notably, they must associate with an F-marked constituent, a point which has not been previously noticed before in the literature. This association with focus supplies the phonological condition for prosodic inversion: expletives must attach to a stressed foot. The important difference between expletives and regular focus adverbs is that they must occur at the left edge of a stress bearing word (or foot, as we know from McCarthy). Thus, they cannot occur before an unstressed preposition like *to* (32c), but instead must occur directly before the noun (32d), even if it is arguably the entire PP which is F-marked. If the preposition can bear stress, as in (32e), then the expletive can attach directly to it.

- (32) a. Fuckin' [JOHN]_F brought a hoagie to class.

- b. John fuckin' [brought a HOAGIE]_F to class.
- c. *John brought a hoagie fuckin' [to class].
- d. John brought a hoagie [to fuckin' CLASS]_F.
- e. John put a hoagie fuckin' [inside his BACKPACK]_F.

Like *only*, *fuckin'* must c-command the F-marked constituent.

There are some differences between expletives and *only*. In particular, expletives seem to be able to attach to root CPs, shown in (33a), in contrast to (31a). Additionally, expletives cannot occur before the complementizer *that* in English, because it is unstressed (33b), though they can occur directly before the stressed subject with broad focus on the entire CP (33c), associated with nuclear stress on the object.

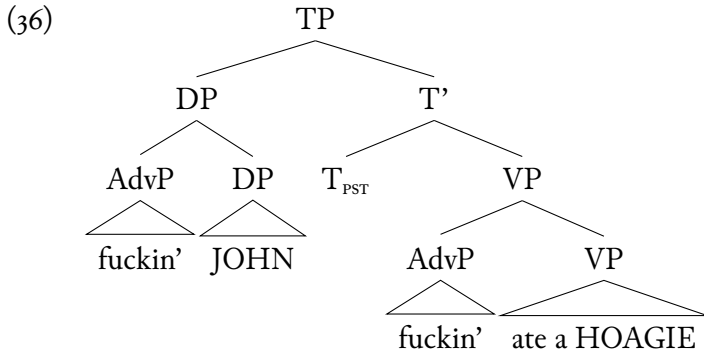
- (33) a. (Q: What happened?) A: Fuckin' [John bought a HOAGIE]_F.
 b. *Mary claimed fuckin' [that John bought a HOAGIE]_F.
 c. (Q: What did Mary claim?) A: (She claimed) [that fuckin' John bought a HOAGIE]_F.

With this observation established, it is plausible to think of cases such as (32d), where the expletive is pronounced inside of a PP, as well as examples such as the following, where an expletive is pronounced inside a subject DP, as resulting from prosodic inversion into the subject DP from a sentence-adjoined position to the first stress-bearing word (34, 35).

- (34) (Q: What happened?) *[Fuckin' [the nice guy I met yesterday] bought a hoagie.]_F
 (35) (Q: What happened?) [[The fuckin' nice guy I met yesterday] bought a hoagie.]_F

There is a clear mismatch between the semantic scope of the expletive over the entire CP and its syntactic position inside of the subject DP. This mirrors nearly exactly the patterns seen in the earlier sections for German and Serbo-Croatian, where prosodic inversion placed clitics inside of syntactic constituents. Unlike German *aber* and Serbo-Croatian clitics, however, English expletives are proclitics on a stressed foot, rather than enclitics on a prosodic word or phrase.

Based on the discussion above, we propose that expressive adverbs *always* originate as adjuncts to major categories, whether DP, PP, CP, or VP.



To account for the optionality of inversion, i.e., the fact that often prosodic inversion does not always apply, we propose there are distinct entries for the relevant expletives. Specifically, we suggest that there is one lexical entry which does not *P*-select for nuclear stress but only for prosodic words, allowing it to attach to unstressed verbs such as *ate* above (36). The requirement

that it must attach to a phonological word accounts for its local displacement across prepositions or complementizers as we observed above. A second lexical entry is the infixing variant, which adjoins to any major syntactic category, but $\mathcal{P}$ -selects for a foot bearing stress,⁸ triggering prosodic inversion inside of words; i.e., infixation (37b).

- (37) a. Vocabulary Item for $f\lambda kin$
- $$[\text{ADV}, \text{EXPR}] \longleftrightarrow \begin{cases} \mathcal{F}_3 : & /f\lambda kin/ \\ \mathcal{P}_3 : & X_{-[\omega]} \\ \mathcal{R}_3 : & \emptyset \end{cases}$$
- b. Vocabulary Item for $f\lambda kin=$
- $$[\text{ADV}, \text{EXPR}] \longleftrightarrow \begin{cases} \mathcal{F}_4 : & /f\lambda kin/ \\ \mathcal{P}_4 : & X_{-[\text{f} \dots ' \sigma \dots]} \\ \mathcal{R}_4 : & \emptyset \end{cases}$$

These two vocabulary items might have slightly different semantics: *fuckin'* in (37a) has sentence-level scope, expressing excitement or negative affect towards the entire proposition, while the proclitic associates the heightened emotion with the focused VP, DP, or lexical item, the expressive value narrowly focused on the choice of alternative relative to some larger question under discussion.

Like Serbo-Croatian clitics, the prosodic subcategorization of the expletive proclitic is inviolable; expletives demand a host. One piece of evidence for this claim is the existence of dummy hosts for *fuckin'*=, and its oddness in isolation:

- (38) a. Fuckin'-A
b. *Fuckin'

Below we provide a sample derivation where proclitic *fuckin'*= originates in a VP-adjoined position before it is placed before the nuclear stress located in the final foot of the phonological phrase to which it attaches. Like a focus-associating adverb, this position can be associated either with broad VP focus (going to Acapulco rather than doing something else), or with narrow focus on the goal: going to Acapulco rather than *somewhere* else). It so happens that in such an example the foot bearing primary stress within this constituent is inside of a word. This results in apparent infixation, as exponent insertion itself applies to prosodic representations, and hence is blind to morphosyntactic boundaries.

- (39) a.
-
- b. Linear concatenation: $[\text{ADV}, \text{EXPR}] \frown (\phi \text{ going}_{\omega} (\omega \text{ to (Aca)}('pulco)))$

⁸Yu (2026) identifies several naturally occurring examples of expletives attaching to underlyingly unstressed syllables which then must be pronounced with stress, such as the exclamation 'guilt-freaking-tee' from *The Cable Guy*. Such point towards adopting a constraint-based implementation of Prosodic Inversion, where one way of satisfying the subcategorization constraints specified by the expletive is by inserting a stress on its host to the right, an impossible derivation if prosodic inversion is always determined by the input stress. The challenge is determining and modeling where such exceptional cases would be allowed.

- c. Exponent choice (=VI): $([ADV,EXPR] \leftrightarrow \mathcal{F}_4, \mathcal{P}_4, \mathcal{R}_4) \frown (\phi \text{ going}_{\omega} (\omega \text{ to } ({}_f Aca)({}_f 'pulco)))$
- d. Exponent insertion (+infixation): $(\phi \text{ going}_{\omega} (\omega \text{ to } ({}_f Aca)({}_f fuckin=)({}_f 'pulco)))$

One final question is how to analyze the cases of optionality in (28); we argue that these always arise due to optionality between two different prosodic parses of the word. This optionality arises due to the Stray Syllable Adjunction principle proposed by Liberman & Prince (1977), a process which repairs unparsed material on any metrical level. As summarized by (Hayes 1995: p. 108), “once a given layer of metrical structure has been created, any element that is **stray** on that layer is incorporated as a metrically weak member of an adjacent constituent” (bolding maintained from source). Via stray adjunction, the [ma] of *tatamagouchi* could sometimes be adjoined to the first and other times the second foot, resulting in two different valid positions of *fuckin’*.

- (40) a. $(,ta.ta)(ma.('gou.che)) \rightarrow (,ta.ta)(fuckin=)(ma.('gou.che))$
- b. $(,ta.ta.ma)('gou.che) \rightarrow (,ta.ta.ma)(fuckin=)('gou.che)$

Prosodic inversion applies identically in the two cases, satisfying the selectional requirements of the expletive.

Expletive infixation is a useful case study because it represents an intermediate case between infixation and other cases of prosodic inversion. Like infixation, it interrupts words, but like German prosodic inversion of *aber*, it can apply across several words. These commonalities highlight the advantages of a unified model of the two phenomena.

7 Discussion and implications

In this section we discuss several implications of our proposal, first discussing the prospects for a unified analysis, then examining issues of minimal shift, phase containment, and last the predicted typology of infix and clitic placement.

7.1 Unifying locality conditions on clitics and infixes

The crux of our proposal is this: clitics and infixes are both just vocabulary items with prosodic subcategorization requirements; they themselves can be words or subwords. We take as the null hypothesis that any category on the prosodic hierarchy can be the target of selection via subcategorization requirements. In Cophonologies by Phrase, both are modeled in a single, unified way, as $\mathcal{P}$ -specifications in their lexical entry. Traditional ‘infixes’ have $\mathcal{P}$ -specifications that target sub-word constituents (feet, syllables, segments). On the other hand, traditional ‘clitics’ have $\mathcal{P}$ -specifications that target prosodic constituents at the word-level or higher (words, phrases, intonational phrases), such that they are word- or phrase-peripheral. We have already seen that some clitics look like words, such as German *aber*, while others are canonically weak clitics, not having their own stress, for example, as in the Serbo-Croatian inflectional clitics discussed in the previous section. Conversely, while canonical infixes are affixes, English expletive infixes are essentially words which select for positions inside of other. Thus, it seems, both words and affixes can select for positions at all levels of the prosodic hierarchy, as predicted by a unified account.

Anderson (2005: 143) argues against a strictly unified treatment of infixes and clitics, saying, “We certainly do not want clitic positioning to be determined in the same way [as infixation],

[...] it is virtually never the case that clitics appear in a position where they are properly included within a preexisting PWord.”

On the contrary, however, we argue that we *do*, in fact, want the position of clitics and infixes to be determined in the same way. English expletive infixation is just one counter example to Anderson’s observation: English expletives show that prosodic inversion *can* break up an existing PWord. Similarly, we believe that this is likely also the correct analysis of *endoclititics* in Nakh-Daghestanian languages like Udi (Harris 2002) or Andi (Maisak 2021).

Halpern (1995: 68) also argues against a unified treatment of infixation and prosodic inversion, claiming that there are no instances of observing that no cases of prosodic inversion result in proclisis, infixes do select for the left edge of specific phonological category. However, expletive infixation provides a counterexample to this claim by Halpern, since it prosodically inverts to the *beginning* of a foot containing nuclear stress.

As noted by Halpern as well as by Weisser (2024), other cases of selection for the left edge of prosodic constituents are simply unlikely to produce prosodic inversion. For example, if a head selects for the left edge of a prosodic word or phrase in a head-initial language, this will simply result in that head being phonologically dependent on the following material. However, there are some cases where heads in head-final languages seem to unexpectedly appear as prefixes, for example agreement morphemes in Cupeño and Navajo, both discussed in Harley (2011).

7.2 Minimal compliance

A property which is characteristic of prosodic inversion, exemplified both by *aber* and Serbo-Croatian second-position clitics, is that they identify the closest available edge from their base position. This property is similar to the kind of minimality effect which has been observed to be a general property of long-distance dependencies in syntax since at least Rizzi (1990). For example, both A- and A'-movement target the closest eligible phrase to the movement site. Similarly, Weisser (2024) shows that shifting coordinators only ever target the first phonological word or phonological phrase to their right when their position is prosodically determined.

This kind of minimality distribution also holds for infixation, though it is a much less celebrated feature of it: infixation always targets phonological hosts at the left or right edge of a word, it does not skip. The fact that minimal compliance is shared by prosodic inversion and infixation provides a further argument that their formal analysis is due to the same mechanism.

However, nothing about our analysis derives this generalization. In particular, the requirement that German *aber* follow a phonological phrase would be satisfied if moved to a position following the second prosodic phrase in the second conjunct, similarly the requirement that Serbo-Croatian inflectional clitics must follow a prosodic word would be satisfied if they were moved to a position following the second prosodic word. What rules these derivations out?

There are three options. The simplest solution would be to adopt an economy principle which explicitly ruled out needless violations of input linear order, such as the following:

(41) *Order preservation*

Don’t make unnecessary changes to the linear order.

This kind of global economy principle was adopted widely early in the Minimalist Program. For example the principle Attract Closest is conceived as an interface condition which rules out

derivations where a constituent moves farther than necessary, or where the closest eligible constituent bearing a relevant feature does not move Chomsky (1995). Such a constraint could be adopted as essentially banning instances of exponent insertion which violated input linear orders.

The second option would be to treat (41) as a violable constraint, such as the constraint *NO SHIFT* adopted by Bennett et al. (2016). In such an analysis, mapping from syntax to phonology could be modeled with a round of constraint satisfaction specifically tasked with the job of generating a prosodic representation from the syntactic tree. This describes the assumptions of Match Theory (Selkirk 2011) and work building on it (Bennett et al. 2016, Elfner 2015). In such a view, prosodic inversion occurs when *NO SHIFT* or an equivalent constraint was outranked by a constraint enforcing the *P*-specification, either a general constraint enforcing subcategorization requirements, such as the *SATISFY* constraint proposed by Grimshaw (1997), or by interpreting *P*-specifications as indexed alignment constraints, as in Anderson (2005). Under such analyses, minimality effects arise simply because any derivation with a non-minimal violation of *NO SHIFT* will be harmonically bound by those with the minimal violations necessary to satisfy the prosodic subcategorization requirements of the relevant lexical item.

The final analytic approach to enforce the minimality generalization would be to derive minimality as a by-product of how a morphological rule which derives prosodic inversion is characterized. One approach would be to define the environment for this rule with linear adjacency, where the inverting morpheme only inverts with a prosodic constituent it is linearly adjacent to. This is essentially the approach of Marantz (1988), Halpern (1995), Embick & Noyer (2001), and in its most general formulation, Weisser (2024), who calls this rule *Integration*, resulting in the reordering and rebracketing of a preceding morpheme or phrase to the right edge of the phonological word or phrase following it.

One outstanding question about this approach is that it attempts to derive the particular inverting behavior of a clitic based on its position in the tree or prosodic hierarchy rather than as a by product of a subcategorization feature. It is not clear this is desirable; infixation requires specific features which are not predictable from the shape of the infix, so to with various cases of prosodic inversion, in particular the case of expletive infixation, which does not target an adjacent word or phrase at all. As such, rules based strictly on linear adjacency are generally incompatible with the view that all instances of prosodic inversion and infixation are ultimately driven by the prosodic subcategorization features of their host.

There are other rule-based perspectives which have not been explored in the literature, such as a search-based inversion algorithm akin to syntactic Agree.

Ultimately the decision between these different formulations of prosodic inversion will depend on the full range of attested cases. On one hand, if specific patterns of inversion can be predicted from their syntactic properties (e.g. all expletives target nuclear stress, etc, sentence-level adjuncts invert with phonological phrases, phrasal adjuncts with words, etc.), a less general account than the one advocated for here may be preferred. However, if a diversity of words and morphemes are eventually found exhibiting the full range of subcategorization requirements, this will constitute good evidence that prosodic inversion and infixation should both be driven by subcategorization features. Below we offer a preliminary exploration of this typology.

7.3 Phase-containment predictions

Sande et al. (2020) propose the phase containment principle, which follows from applying the

morpheme-specific constraint ranking $\mathcal{R}$ of CbP-style vocabulary items in a spell-out-by-phase approach (42).

(42) *The Phase Containment Principle:*

Morpheme-specific phonological operations conditioned internal to a phase cannot affect the phonology of phases that are not yet spelled out.

Here we extend the phase containment principle to encompass not just $\mathcal{R}$ specifications but also the predictions of applying morpheme-specific prosodic subcategorization requirements $\mathcal{P}$ in a spell-out-by-phase approach (43).

(43) *The Revised Phase Containment Principle:*

Morpheme-specific phonological requirements ($\mathcal{P}$, $\mathcal{R}$) cannot access phases that are not yet spelled out.

An implication of (43) is that a morpheme with a $\mathcal{P}$ -specification can target a prosodic element within the same or a previously spelled out phase domain, but not one introduced in a higher syntactic phase.

Weisser (2024) in fact argues for a very similar conclusion based on the observation that monosyndetic coordinators only prosodically invert with the conjunct on their right, which he assumes has been previously spelled out, and never those on their left.

7.4 The predicted typology of infix and clitic placement

Our model predicts that morphemes could target a position adjacent to any element on the prosodic hierarchy. There are existing typologies of infixation (Kalin 2022, Moravcsik 1977, Yu 2003, 2007), and cliticization (Anderson 2005, Klavans 1980, 1985, Zwicky 1977). Based on these existing typologies and our own additional research, we have filled in the predicted typology of prosodic subcategorization targets in Table 1. As we predict, all of the predicted targets are attested in at least one known language. See Yu (2003) for a more fine-grained typology of sub- ω targets than the one provided here.

Table 1: Predicted typology of infix and clitic targets

	Prefix	Suffix
C	SiSwati (Bantu; (Yu 2003: 17))	Mlabri (Mon-Khmer; (Yu 2003: 12))
V	Toratan (Austronesian; (Yu 2003: 19))	Alabama (Muskogean; (Yu 2003: 25))
σ	KiChaga (Bantu; (Yu 2003: 32))	Chamorro (Austronesian; (Yu 2003: 43))
Σ	English expletive infixation	Ulwa (Misumalpan; (Yu 2003: 41-42))
ω	Navajo verbal prefixes (Athabaskan; (Harley 2011))	Serbo-Croatian 2P clitics
ϕ	Guébie possessive pronouns (Kru; (Sande 2017))	German <i>aber</i> (Weisser 2020)

We additionally predict that intonational phrases and utterances should be targetable by $\mathcal{P}$ (i.e., clause-final negation in Lobi (Gur; Sande’s fieldnotes)). Our approach does not prevent a traditional ‘clitic’ from targeting a sub-word constituent on the prosodic hierarchy. We think this is a good prediction, first, because expletive infixation targets a subword constituent—a foot—and second, because of endoclititics. We expect that other elements that can attach to multiple morphosyntactic categories (i.e., clitics) may also target segments or syllables.

In our view, then, there is no reason to draw a line between what have traditionally been called ‘infixes’ on the one hand and ‘clitics’ on the other; they are both just morphemes with prosodic subcategorization requirements.

8 Conclusions

Clitics and infixes are both morphemes that contain lexically-encoded subcategorization features for particular prosodic units. Cophonologies by Phase offers a natural way to account for both clitic and infix position, via the $\mathcal{P}$, or prosodic requirements of individual lexical items. This talk represents the first exposition of the functions of $\mathcal{P}$ in Cophonologies by Phase.

Along the way, we have also proposed a specific implementation of conditions on insertion (COINs) and conditions are position (COPs) (Kalin & Rolle 2022): COINs apply to the whole vocabulary item, while COPs are part of the phonological content of the VI, namely, $\mathcal{P}$.

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Abbreviations

ADJ = adjective, AUX = auxiliary verb, CONJ = conjunction, FOC = focus,

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